

A Universal Constant Two for Derivations on In-Tree Tensor Algebras

Candidate full answer to Conjecture 6.27 of arXiv:2507.22508

26 August 2026

Status. Candidate full solution, likely valid, pending human review. For every finite in-tree G and every derivation δ on its tensor algebra $\mathcal{A}_G$, there is $S \in \mathcal{A}_G$ such that $\delta = \delta_S$ and $\|S\| \leq 2\|\delta\|$. Thus the conjectured universal constant exists, with $C = 2$.

1 Source conjecture

Huang and Ma ask for a graph-independent implementation constant for derivations on tensor algebras of finite in-trees [1, Conjecture 6.27]. Here an in-tree is a finite directed tree with a range root v_0 such that there is a directed path from every vertex to v_0 .

$$\|T\|^2 = \max_{1 \leq j \leq n} |s_j|^2 + |t_j|^2. \quad (6.1)$$

Let $A = aL_{v_j} + bL_{e_j}$. Then $\delta(A) = (bs_j - at_j)L_{e_j}$. It follows that

$$|bs_j - at_j|^2 = \|\delta(A)\|^2 \leq \|\delta\|^2 \|A\|^2 = \|\delta\|^2 (|a|^2 + |b|^2).$$

Let $a = -\overline{t_j}$ and $b = \overline{s_j}$. Then $|s_j|^2 + |t_j|^2 \leq \|\delta\|^2$. Hence $\|T\| \leq \|\delta\|$ by (6.1). This completes the proof. $\square$

At the end of this subsection, we propose the following conjecture.

Conjecture 6.27. *Suppose G is an in-tree. Then for every derivation δ on $\mathcal{A}_G$, there exists $T \in \mathcal{A}_G$ such that $\delta = \delta_T$ and $\|T\| \leq C\|\delta\|$, where $C \geq 1$ is a universal constant.*

Figure 1: Conjecture 6.27 in the source, arXiv PDF page 33.

With the source convention, $\delta_T(A) = AT - TA$. The sign convention is irrelevant to the norm estimates below. The source proves that every derivation on $\mathcal{A}_G$ is inner when G is a finite tree [1, Lemma 6.2]; the open point is a bound independent of the number and shape of the vertices.

2 Proof intuition

Start with any implementer T . Averaging T over the diagonal sign unitaries separates it into its vertex-diagonal part D and its positive-length path part N . The averaging identity writes N as an average of values of the derivation on norm-one diagonal unitaries, so $\|N\| \leq \|\delta\|$.

The diagonal coefficients are controlled without adding estimates along edges. For a whole directed path p , the L_p coefficient of $\delta(L_p)$ is exactly the difference of the diagonal coefficients at the two endpoints. Applying this to the path from each vertex directly to the root bounds the entire diagonal, modulo the root scalar, by $\|\delta\|$. Adding the two independent pieces gives the constant 2.

3 The theorem

For a path p , write $s(p)$ and $r(p)$ for its source and range, and let L_p be the corresponding left creation operator. The vertex projections $(L_v)_{v \in \mathcal{V}(G)}$ are pairwise orthogonal and sum to the identity.

Theorem 1. *Let G be a finite in-tree. For every derivation $\delta : \mathcal{A}_G \rightarrow \mathcal{A}_G$, there exists $S \in \mathcal{A}_G$ such that*

$$\delta = \delta_S, \quad \|S\| \leq 2\|\delta\|.$$

Consequently Conjecture 6.27 holds with the universal constant $C = 2$.

Proof. By the source's Lemma 6.2, choose $T \in \mathcal{A}_G$ with $\delta = \delta_T$. Let

$$\mathcal{D}_G = \text{span}\{L_v : v \in \mathcal{V}(G)\}.$$

For a sign vector $\varepsilon = (\varepsilon_v)_{v \in \mathcal{V}(G)} \in \{-1, 1\}^{\mathcal{V}(G)}$, put

$$U_\varepsilon = \sum_{v \in \mathcal{V}(G)} \varepsilon_v L_v.$$

This is a unitary element of $\mathcal{D}_G \subseteq \mathcal{A}_G$. Define the diagonal expectation

$$E_{\mathcal{D}}(X) = 2^{-|\mathcal{V}(G)|} \sum_{\varepsilon} U_\varepsilon X U_\varepsilon^*.$$

If p has positive length, then $r(p) \neq s(p)$ because G is a tree, and

$$U_\varepsilon L_p U_\varepsilon^* = \varepsilon_{r(p)} \varepsilon_{s(p)} L_p.$$

The sign average therefore kills every positive-length path term and fixes every vertex term. Write

$$D = E_{\mathcal{D}}(T) = \sum_{v \in \mathcal{V}(G)} \lambda_v L_v, \quad N = T - D.$$

Since

$$T - U_\varepsilon T U_\varepsilon^* = (T U_\varepsilon - U_\varepsilon T) U_\varepsilon^* = -\delta(U_\varepsilon) U_\varepsilon^*,$$

we obtain

$$\|N\| \leq 2^{-|\mathcal{V}(G)|} \sum_{\varepsilon} \|\delta(U_\varepsilon)\| \leq \|\delta\|. \tag{1}$$

Now fix a positive-length path p . The Fourier coefficient at L_p is the contractive matrix functional

$$\phi_p(X) = \langle X \xi_{s(p)}, \xi_p \rangle.$$

The diagonal part gives

$$\phi_p(L_p D - D L_p) = \lambda_{s(p)} - \lambda_{r(p)}.$$

Every term in $L_p N$ or $N L_p$ is either zero or is supported on a path of length strictly larger than $\ell(p)$. Thus it has zero L_p coefficient, and hence

$$|\lambda_{s(p)} - \lambda_{r(p)}| = |\phi_p(\delta(L_p))| \leq \|\delta(L_p)\| \leq \|\delta\|. \quad (2)$$

Let v_0 be the range root. For each vertex $v \neq v_0$, the in-tree property supplies a directed path p_v with $s(p_v) = v$ and $r(p_v) = v_0$. Applying (2) to this entire path gives

$$|\lambda_v - \lambda_{v_0}| \leq \|\delta\|.$$

Since the L_v are orthogonal projections,

$$\|D - \lambda_{v_0} I\| = \max_{v \in \mathcal{V}(G)} |\lambda_v - \lambda_{v_0}| \leq \|\delta\|. \quad (3)$$

Finally set $S = T - \lambda_{v_0} I$. Subtracting a scalar does not change the inner derivation, while (1) and (3) give

$$\|S\| \leq \|N\| + \|D - \lambda_{v_0} I\| \leq 2\|\delta\|.$$

□

4 Verification, scope, and novelty boundary

The companion script `code/verify_tree_averaging.py` constructs the finite left-regular path matrices for random rooted in-trees and checks the sign-average identity and the exact path-coefficient formula. The computation is diagnostic only; the proof is the finite averaging and coefficient calculation displayed above.

The theorem proves the existence assertion of Conjecture 6.27. It does not assert that $C = 2$ is optimal. The source obtains $C = 1$ for out-trees and for one star-shaped family of in-trees [1, Propositions 6.25–6.26].

Bounded exact-title, arXiv-id, author, exact-conjecture, and keyword searches through 26 August 2026 found the source and mirrors of it, but no later primary source recording this proof or a resolution of Conjecture 6.27. Priority is not certified and expert review remains necessary.

5 Human review recommendation

Check three convention-sensitive points: the formula $U_\varepsilon L_p U_\varepsilon^* = \varepsilon_{r(p)} \varepsilon_{s(p)} L_p$; the assertion that products involving N cannot contribute to the L_p coefficient; and the use of the whole path from a vertex to the root rather than an edge-by-edge sum. Once these are verified, the constant-two estimate follows immediately.

References

- [1] L. Huang and M. Ma, *Free semigroupoid algebras and the first cohomology groups*, arXiv:2507.22508 (2025).